NAM mining: iPSC protocols for retinal organoids, RGCs and photoreceptors (37686457)

OpenScientist deep-research run for the Complement-ARIE disease-paper round (nams-registry#37).

- **Reference:**

  37686457 · DOI 10.3390/ijms241713652 · *Int J Mol Sci* 24(17):13652 (2023) · open access, PMC10487913.
- **Title:** *Establishing Functional Retina in a Dish: Progress and Promises of Induced Pluripotent Stem Cell-Based Retinal Neuron Differentiation* (Wong, Yip, Huang).
- Full text retrieved and parsed via Europe PMC (62,800 chars); every quote below verified against it.
- dismech assessed at commit `3b9cbe3c4b1374a2631c1d924054535dd062d532` (2026-11-13); nams-registry at `38f81296983cbb71c3f8912a09f793836864bb5a`.

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1. Summary — answer to the research question

37686457 is a **methods review**, not a disease study. It describes three canonical, **wild-type** human-iPSC cell products — retinal organoids, retinal ganglion cells (RGCs) and photoreceptors — across published differentiation protocols (Tables 1–3), with no patient-derived or disease-genotype line anywhere in the body. It therefore yields **three canonical NAM records** and **cell-type / substrate model links**, plus unusually good **fetal-retina fidelity** content — but it does **not** support any disease-mechanism *recapitulation* claim on its own. Accordingly, every model-to-mechanism pairing is curated with relationship `MEASURES` (not `RECAPITULATES`), fidelity `MODERATE`, and an explicit `CAUSE_UNREPRESENTED` divergence (the disease lesion is not

encoded). The `retinoid_visual_cycle_disruption` module is confirmed to be the **wrong anchor** (it scopes to RPE chromophore recycling, not differentiation chemistry), and the canonical RGC record **cross-references** the existing isogenic OPTN(E50K) hPSC-RGC instance rather than duplicating it.

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2. What kind of paper this is (verified)

- Reviews **exactly three cell products**, each with its own protocol table: retinal organoids (Table 1, 7 studies), RGCs (Table 2, 9 studies), photoreceptors (Table 3, 9 studies), plus validation methods (IF, qPCR, flow cytometry, scRNA-seq, TEM, patch-clamp, MEA, calcium imaging).
- Named diseases appear **only as motivation**: *"various retinal degenerative diseases such as retinitis pigmentosa (RP), glaucoma, and age-related macular degeneration (AMD) can lead to vision loss or blindness"* (abstract); RP/glaucoma/AMD recur in the Introduction and Perspectives as rationale.
- **No disease genotype or patient line** is described. The scope statement is explicit: *"This review focuses on the protocols, culture conditions, and techniques employed in differentiating retinal neurons from iPSCs."*
- The three products are the cell populations whose loss **defines** the named diseases: photoreceptors (RP, AMD) and RGCs (glaucoma).

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3. Deliverable 1 — Model-to-mechanism rows (one per pairing, quote-verified)

Mechanism nodes are from `kb/modules/photoreceptor_degeneration.yaml`. All relationships are `MEASURES` because the models carry no disease lesion (see §5).

#	Model (cell product)	Mechanism node (module#node)	Relationship	Verbatim support from P 37686457 full text
1	iPSC retinal organoid	`photoreceptor_degeneration#Photoreceptor Gene Defect` (substrate: rod/photoreceptor cell)	MEASURES	*"Mature photoreceptors are identified using specific cell markers such as S opsin (OPN1SW), L/M opsin (OPN1L/MW), RHO, CRX, and NR2E3"* — organoids contain and read out the photoreceptor population the node is situated in.
2	iPSC retinal organoid	`photoreceptor_degeneration#Secondary Cone Degeneration and Outer Retinal Thinning` (substrate: outer nuclear layer / cones)	MEASURES	*"iPSC-derived retinal cells exhibit the temporal order of embryonic retinal development and replicate the cellular mosaics and lamination of the human retina"* (Introduction) — provides the laminated outer-retinal tissue scale; but the disease lesion is absent.
3			MEASURES	

#	Model (cell product)	Mechanism node (module#node)	Relationship	Verbatim support from P 37686457 full text
	iPSC photoreceptor	`photoreceptor_degeneration#Photoreceptor Gene Defect` (substrate: rod/photoreceptor cell)		*"OTX2 and recoverin (RCVRN) are used to demonstrate functional photoreceptors after differentiation … The expression of RHO confirms the presence of functional rod cells"* — reads out the phototransduction components whose mutation is the node's trigger, in healthy form.
4	iPSC photoreceptor	`photoreceptor_degeneration#Photoreceptor Metabolic and Oxidative Stress` (substrate: photoreceptor cell)	MEASURES (maturation/ ultrastructure readout only)	*"TEM analysis has also been employed to compare the maturation of photoreceptors differentiated with 9-cis retinal treatment versus ATRA treatment, showing a more advanced stage of maturation in photoreceptors treated with 9-cis*

#	Model (cell product)	Mechanism node (module#node)	Relationship	Verbatim support from P 37686457 full text
				retinal" — reads out outer-segment ultrastructure/ maturation, the substrate of oxidative-stress pathology, without inducing it.
5	iPSC RGC	*(no photoreceptor_degeneration node applies)* — RGC is the glaucoma substrate; see cross-reference to OPTN record	MEASURES	*"co-expression of BRN3A and SNCG is a characteristic feature of mature RGCs"*; functional readout: *"calcium entry in immature RGCs upon depolarization induced by a GABA agonist … Mature RGCs did not demonstrate a depolarization-induced calcium influx."*

> RGCs do **not** map to a `photoreceptor_degeneration` node. Their disease home is glaucoma; the pairing is handled by cross-reference to the existing `OPTN-related_Open_Angle_Glaucoma.yaml` instance (§6), not by a new mechanism link here.

4. Deliverable 2 — `ModelMechanismLink` drafts (dismech)

These attach as `experimental_models[].modeled_mechanisms` on the disease entries (same structure as the existing OPTN record). Three example links are given; the rest follow the §3 rows. `model_scale: CELLULAR` for the purified cell products; `TISSUE` for the organoid.

4a. On `Inherited_Retinal_Dystrophy.yaml` — photoreceptor product

```
experimental_models:
- name: iPSC-derived photoreceptors (canonical differentiation protocols)
  experimental_model_type: IPSC_DERIVED_MODEL
  cell_source: Human induced pluripotent stem cells; no disease genotype (wild-type reference pr
  culture_system: 2D directed differentiation and organoid-derived/bioreactor formats (review of
  description: >
    Canonical wild-type human iPSC-derived photoreceptors reviewed in <a href="https://pubmed.nc
    (Table 3): signaling-modulator, modulator-free, small-molecule-inhibitor and
    stirred-tank-bioreactor protocols. Models the human photoreceptor cell type and
    its maturation, NOT a disease lesion.
  modeled_mechanisms:
  - target: Photoreceptor Metabolic and Oxidative Stress
    relationship: MEASURES
    fidelity: MODERATE
    model_scale: CELLULAR
    description: >
      Reads out photoreceptor identity, phototransduction-component expression and
      outer-segment ultrastructure/maturation (the substrate of the node), without
      encoding or inducing the oxidative-stress cascade.
    readouts:           # range = ExperimentalReadout; each readout's `target` repeats the link
    - name: Photoreceptor identity marker panel (OTX2/RCVRN/CRX/NRL/NR2E3, RHO/opsins)
      target: Photoreceptor Metabolic and Oxidative Stress
      direction: UNCHANGED      # wild-type cells; healthy expression, no stress phenotype
      assays: [immunofluorescence, qPCR, flow cytometry]
      interpretation: >
        Confirms the phototransduction/identity substrate is present and healthy; it
        reads out the node's cell type, not an oxidative-stress phenotype.
    - name: Photoreceptor outer-segment ultrastructure (TEM)
      target: Photoreceptor Metabolic and Oxidative Stress
      direction: ALTERED        # maturation advances with 9-cis-retinal vs ATRA
      assays: [transmission electron microscopy]
      interpretation: >
        TEM of outer segment, inner segment, connecting cilium and basal body; 9-cis-retinal
        yields "a more advanced stage of maturation" than ATRA — a maturation readout, not disea
    limitations: >
      Wild-type cells with no disease allele; developmentally immature relative to
      adult photoreceptors; scRNA-seq shows photoreceptor developmental trajectories
      that diverge from fetal retina.
    divergences:
    - divergence_type: CAUSE_UNREPRESENTED
      description: >
        No retinal-dystrophy lesion (e.g. RHO, PDE6, RPGR, PRPH2, RPE65, ABCA4) is
        present; the model is the healthy cell type, so degeneration is not reproduced,
        only its cellular substrate is measured.
    - divergence_type: CALIBRATION_PROVENANCE
      description: >
        Maturation benchmarked against human fetal (not adult) retina by scRNA-seq,
        and photoreceptor trajectory is the cell type the same comparison flags as divergent.
```

```
    evidence:
    - reference: <a href="https://pubmed.ncbi.nlm.nih.gov/37686457/" rel="noopener noreferrer" t
      reference_title: "Establishing Functional Retina in a Dish: Progress and Promises of Induc
      supports: SUPPORT
      evidence_source: OTHER
      snippet: "TEM analysis has also been employed to compare the maturation of photoreceptors
      explanation: >
        A review (evidence_source OTHER) documenting the TEM maturation readout the
        link records, with no disease lesion involved.
```

4b. On `Age_Related_Macular_Degeneration.yaml` — organoid product

Same shape, `target: Secondary Cone Degeneration and Outer Retinal Thinning`, `model_scale: TISSUE`, readouts = laminar/outer-nuclear-layer markers + scRNA-seq composition, with an additional divergence:

```
    divergences:
    - divergence_type: CAUSE_UNREPRESENTED
      description: No AMD-associated lesion (e.g. complement/CFH, drusen biology) is encoded.
    - divergence_type: BOUNDARY_OMISSION
      description: >
        No RPE–photoreceptor–choriocapillaris–immune interface; the review notes
        multifactorial AMD "might require … co-culturing with RPE or the incorporation
        of the vascular network" to recapitulate physiological interactions.
```

Supporting snippet (verified): *"regarding multifactorial traits such as AMD or diabetic retinopathy, more complicated culture conditions like co-culturing with RPE or the incorporation of the vascular network might be required to recapitulate physiological interactions."*

4c. On `Glaucoma.yaml` — RGC product

`experimental_model_type: IPSC_DERIVED_MODEL`; **no `photoreceptor_degeneration` target** (wrong cell type). The RGC product is listed as a glaucoma `experimental_model` with a narrative cross-reference note to the dismech OPTN(E50K) instance (§6; not a registry `compared_with`, which must point at an nams-registry record). If a glaucoma RGC pathophysiology node exists, attach `relationship: MEASURES`, readouts = BRN3A/SNCG/RBPMS/THY1 identity + patch-clamp + GABA-agonist calcium maturation staging; `CAUSE_UNREPRESENTED` divergence (no OPTN/MYOC/elevated-IOP lesion).

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5. Deliverable 3 — Three NAM records (nams-registry)

Augment-vs-create: `nams/retinal-organoid-pluripotent-stem-cell.yaml` **already exists** (sourced from the STEMCELL/LSCN-2022 resource; Eiraku 2011 + Zhu 2013). The correct action is to **augment** it with this paper; the **RGC and photoreceptor records are new**.

Granularity verdict (per round rule, one record per product): the protocol variants in Tables 1–3 are **described differences on one record each**, not separate records — *except* the **NGN2-overexpression RGC protocol** (Luo et al., ref 58), which is transcription-factor forward-programming (*"overexpression of neurogenin 2 (NGN2) through lentiviral transduction to expedite the differentiation process"*, *"completed in approximately one week"*) rather than stepwise directed differentiation. It is the **one borderline case** that could merit its own record, and notably it is the very protocol whose cells the paper reports as *more* fetal-like than organoid RGCs. **Recommendation: keep it as a described variant but flag it in the round's granularity thread for a ruling.**

> Provisional NAMO typing (as the `sh-sy5y` records do): organoid = `Organoid`; RGC/photoreceptor purified cultures = `TwoDCellCulture` (direct differentiation) with an organoid-dissociation route noted — flag to maintainers, since these products are produced both as 2D cultures and by organoid dissociation.

5a. AUGMENT `nams/retinal-organoid-pluripotent-stem-cell.yaml`

Add to the existing record:

```
references:              # append
- id: doi:10.3390/ijms241713652
  title: 'Establishing Functional Retina in a Dish: Progress and Promises of Induced Pluripotent
  authors: [Wong IYH, Yip WWK, Huang WX]   # verify author initials against record style
  journal: International Journal of Molecular Sciences
  year: 2023
applications:            # append
- description: >-
    In vitro modeling of retinal neuron differentiation and retinal disease, and a
    source of photoreceptors/RGCs for replacement-therapy research. scRNA-seq shows
    organoids and retinospheres "closely resembled the fetal retina in terms of
    cellular composition and gene expression patterns".
  source_urls: [https://doi.org/10.3390/ijms241713652]
  supporting_references: [doi:10.3390/ijms241713652]
limitations:            # add
- description: >-
    "some differences were observed in the developmental trajectories of specific
    cell types, such as photoreceptors, between the organoids and the fetal retina" —
    composition is fetal-like but trajectory is not fully matched; organoids are
    developmentally immature.
  source_urls: [https://doi.org/10.3390/ijms241713652]
  supporting_references: [doi:10.3390/ijms241713652]
- description: >-
    Multifactorial disease modeling (AMD, diabetic retinopathy) "might require …
    co-culturing with RPE or the incorporation of the vascular network"; a bare
    organoid omits the RPE/vascular/immune interface.
  source_urls: [https://doi.org/10.3390/ijms241713652]
  supporting_references: [doi:10.3390/ijms241713652]
source_resources:       # append both
- resource: scientific-literature
  source_urls: [https://doi.org/10.3390/ijms241713652]
  accessed_on: '2026-11-14'
- resource: complement-arie-landscape-dashboard
  source_urls: [https://doi.org/10.3390/ijms241713652]
  accessed_on: '2026-11-14'
dismech_status:
  processed_through_dismech: true
  present_in_dismech: false
  dismech_commit: 3b9cbe3c4b1374a2631c1d924054535dd062d532
  assessed_on: '2026-11-14'
last_modified_date: '2026-11-14'
```

5b. NEW `nams/retinal-ganglion-cell-pluripotent-stem-cell.yaml`

```
model:
  id: retinal-ganglion-cell-pluripotent-stem-cell
  type: TwoDCellCulture      # PROVISIONAL; also produced by organoid dissociation. Flag to mair
  name: iPSC-derived retinal ganglion cells
  description: >-
    Human iPSC-derived retinal ganglion cells (RGCs), the inner-retina projection
    neuron lost in glaucoma, reviewed in <a href="https://pubmed.ncbi.nlm.nih.gov/37686457/" rel
    defined conditions, biomaterial scaffolds for neurite outgrowth, dual-SMAD plus
    Wnt inhibition, organoid dissociation with THY1 magnetic-activated cell sorting
    (Rabesandratana et al.), and a rapid ~1-week NGN2-overexpression protocol (Luo
    et al.). Wild-type reference cells; no disease genotype. Directly differentiated
    RGCs resemble fetal RGCs more closely than organoid-derived RGCs.
  organ_modeled: {id: UBERON:0000966, name: retina}
  cell_types: [{id: CL:0000740, name: retinal ganglion cell}]
  cell_source: Human induced pluripotent / pluripotent stem cells; no disease genotype.
  three_d_architecture: MONOLAYER   # PROVISIONAL; organoid-dissociation route is 3D-derived
  culture_conditions: >-
    Multiple protocols: chemically defined medium; dual-SMAD + Wnt inhibition;
    biomaterial scaffolds; organoid dissociation + poly-D-lysine/laminin replating +
    THY1 MACS enrichment; NGN2 lentiviral forward-programming with doxycycline
    induction, GDNF, and Notch inhibition (DAPT).
  references:
  - {id: doi:10.3390/ijms241713652, title: 'Establishing Functional Retina in a Dish …', journal
synonyms: [iPSC-RGC, hPSC-derived retinal ganglion cells, stem-cell-derived RGCs]
species: [{id: NCBITaxon:9606, name: Homo sapiens}]
applications:
- description: >-
    Glaucoma / optic-neuropathy disease modeling and RGC replacement research; a
    platform for RGC electrophysiology (patch-clamp, MEA) and maturation staging.
  source_urls: [https://doi.org/10.3390/ijms241713652]
  supporting_references: [doi:10.3390/ijms241713652]
- description: >-
    Maturation staging by calcium imaging: "calcium entry in immature RGCs upon
    depolarization induced by a GABA agonist … Mature RGCs did not demonstrate a
    depolarization-induced calcium influx."
  source_urls: [https://doi.org/10.3390/ijms241713652]
  supporting_references: [doi:10.3390/ijms241713652]
limitations:
- description: >-
    Organoid-derived and directly-differentiated RGCs differ in identity: directly
    differentiated/fetal RGCs "predominantly expressed BRN3A, whereas most organoid
    RGCs express BRN3B"; the sourcing route changes the product.
  source_urls: [https://doi.org/10.3390/ijms241713652]
  supporting_references: [doi:10.3390/ijms241713652]
- description: >-
    Wild-type reference cells with no glaucoma lesion (OPTN/MYOC/elevated IOP), and
    developmentally immature relative to the adult neurons that degenerate in glaucoma.
  source_urls: [https://doi.org/10.3390/ijms241713652]
```

```
  supporting_references: [doi:10.3390/ijms241713652]
source_resources:
- {resource: scientific-literature, source_urls: [https://doi.org/10.3390/ijms241713652], access
- {resource: complement-arie-landscape-dashboard, source_urls: [https://doi.org/10.3390/ijms2417
compared_with: []
# NOTE: `compared_with.nam` MUST reference an existing nams-registry record. The
# OPTN(E50K) genotype instance (<a href="https://pubmed.ncbi.nlm.nih.gov/39425218/" rel="noopene
# experimental_model on OPTN-related_Open_Angle_Glaucoma.yaml, NOT in nams-registry,
# so it cannot be a compared_with target here. That canonical-vs-instance
# cross-reference is handled on the dismech side (§4c, §6). If a registry record is
# ever created for the OPTN(E50K) instance, add it to compared_with then.
dismech_status:
  processed_through_dismech: true
  present_in_dismech: false
  dismech_commit: 3b9cbe3c4b1374a2631c1d924054535dd062d532
  assessed_on: '2026-11-14'
creation_date: '2026-11-14'
last_modified_date: '2026-11-14'
```

5c. NEW `nams/photoreceptor-pluripotent-stem-cell.yaml`

```
model:
  id: photoreceptor-pluripotent-stem-cell
  type: TwoDCellCulture      # PROVISIONAL; also produced in organoid/bioreactor formats
  name: iPSC-derived photoreceptors
  description: >-
    Human iPSC-derived photoreceptors (rods and cones), the outer-retina cells lost
    in retinitis pigmentosa and AMD, reviewed in <a href="https://pubmed.ncbi.nlm.nih.gov/37686
    signaling-modulator, modulator-free, small-molecule-inhibitor and stirred-tank
    bioreactor protocols. Wild-type reference cells; no disease genotype.
  organ_modeled: {id: UBERON:0000966, name: retina}
  cell_types:
  - {id: CL:0000604, name: retinal rod cell}
  - {id: CL:0000573, name: retinal cone cell}
  - {id: CL:0000210, name: photoreceptor cell}
  cell_source: Human induced pluripotent / pluripotent stem cells; no disease genotype.
  three_d_architecture: MONOLAYER
  culture_conditions: >-
    Multiple protocols: defined conditions with signaling modulators; without
    extrinsic modulators; small-molecule inhibitors; stirred-tank bioreactor
    (Ovando-Roche et al.) for highest yield, enhanced proliferation and reduced
    apoptosis; 9-cis-retinal to advance rod outer-segment maturation.
  references:
  - {id: doi:10.3390/ijms241713652, title: 'Establishing Functional Retina in a Dish …', journal
synonyms: [iPSC-photoreceptors, hPSC-derived photoreceptors, iPSC-derived rods and cones]
species: [{id: NCBITaxon:9606, name: Homo sapiens}]
applications:
- description: >-
    Retinitis pigmentosa / AMD modeling and photoreceptor replacement research;
    ultrastructural validation by TEM (outer segment, inner segment, connecting
    cilium, basal body) and phototransduction-marker readouts.
  source_urls: [https://doi.org/10.3390/ijms241713652]
  supporting_references: [doi:10.3390/ijms241713652]
limitations:
- description: >-
    scRNA-seq flags photoreceptors as a cell type whose "developmental trajectories"
    diverge from fetal retina even when overall composition is fetal-like; cells are
    developmentally immature.
  source_urls: [https://doi.org/10.3390/ijms241713652]
  supporting_references: [doi:10.3390/ijms241713652]
- description: >-
    Wild-type reference cells with no RP/AMD lesion; maturation is protocol-dependent
    (9-cis-retinal advances maturation over ATRA by TEM).
  source_urls: [https://doi.org/10.3390/ijms241713652]
  supporting_references: [doi:10.3390/ijms241713652]
source_resources:
- {resource: scientific-literature, source_urls: [https://doi.org/10.3390/ijms241713652], access
- {resource: complement-arie-landscape-dashboard, source_urls: [https://doi.org/10.3390/ijms2417
dismech_status:
```

```
  processed_through_dismech: true
  present_in_dismech: false
  dismech_commit: 3b9cbe3c4b1374a2631c1d924054535dd062d532
  assessed_on: '2026-11-14'
creation_date: '2026-11-14'
last_modified_date: '2026-11-14'
```

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6. Existing records this sits beside (read before writing)

- **`OPTN-related_Open_Angle_Glaucoma.yaml`** already holds an `experimental_models` entry: *"Isogenic OPTN(E50K) human pluripotent stem cell-derived retinal ganglion cells"* (`IPSC_DERIVED_MODEL`), `modeled_mechanisms: RECAPITULATES` → *"Impaired Autophagic Flux in Retinal Ganglion Cells"*, **fidelity HIGH**, evidence

  P 39425218

  Its own `limitations` already note *"hPSC-derived retinal ganglion cells are developmentally immature relative to the adult neurons that degenerate."* **The canonical RGC NAM record is the baseline this dismech instance is built on → cross-reference it, do not duplicate.** Note on mechanics: this cross-reference is a *narrative/evidence* link, NOT a registry `compared_with` field — `compared_with.nam` can only point at another existing *nams-registry* record, and the OPTN(E50K) instance lives in dismech, not the registry. So the new canonical RGC record leaves `compared_with: []` (§5b), and the OPTN↔canonical relationship is documented here and on the dismech side (§4c). The contrast (RECAPITULATES/HIGH for the genotype instance vs MEASURES/MODERATE for the canonical product) is exactly the nams-registry#86 canonical-vs-instance distinction working as intended.
- `Leber_Congenital_Amaurosis_10.yaml` (CEP290 patient iPSC optic cups/RPE + sepofarsen), `RDH12-Related_Dominant_Retinopathy.yaml` (RDH12 patient organoids) and `Stargardt_Disease.yaml` (patient organoid + STGD1 iPSC-RPE) are all genotype instances built on these same baseline protocols; the canonical records give them a shared, citable baseline.

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7. Deliverable 4 — Review-log line & gap issues

Line for `projects/COMPLEMENT_ARIE_DISEASE_PAPERS.md` (this paper's row):

> P 37686457 — methods review, three canonical iPSC products. **nams-registry:** augmented `nams/retinal-organoid-pluripotent-stem-cell.yaml` (added this paper, fetal-retina fidelity/limitations, `scientific-literature` + `complement-arie-landscape-dashboard`, `dismech_status`); added NEW `retinal-ganglion-cell-pluripotent-stem-cell` and `photoreceptor-pluripotent-stem-cell`. **dismech:** proposed `experimental_models` (MEASURES / MODERATE / CAUSE_UNREPRESENTED) as first models on `Inherited_Retinal_Dystrophy`, `AMD` and `Glaucoma`; RGC cross-referenced to the OPTN(E50K) instance; **no** link to `retinoid_visual_cycle_disruption` (wrong anchor — RA/9-cis here is differentiation chemistry, not RPE chromophore recycling). Granularity: protocol variants are described differences, NGN2 forward-programming flagged as the one borderline case.

Gap issues to open: 1. *No models on three named disease entries* — `AMD.yaml`, `Glaucoma.yaml`, `Inherited_Retinal_Dystrophy.yaml` have no `experimental_models`/`animal_models`/`computational_models`; these canonical products would be the first. 2. *Review-vs-primary citation* — each protocol belongs to a primary paper the review cites; precedent `Parkinsons_Disease.yaml` cites a review (P 42727800). **Done here:** records cite the review (P 37686457) as the source that groups the products, with primary authors named in-text (Sridhar, Luo, Rabesandratana, Ovando-Roche, Reichman); recommend adding primary DOIs as additional `references` where a specific protocol's data is used. 3. *NAMO typing for purified iPSC-derived neurons* (RGC, photoreceptor) produced both as 2D cultures and by organoid dissociation — provisional `TwoDCellCulture`; confirm. 4. *NGN2 forward-programming granularity* — ruling on whether it deserves its own record.

8. Hypotheses: supported / refuted

- **Supported:** Paper is a canonical methods review → three NAM records + fidelity content (fetal-retina scRNA-seq, BRN3A/BRN3B, 9-cis-retinal TEM, GABA-calcium).
- **Supported:** Relationship is MEASURES, not RECAPITULATES — no disease lesion is encoded (schema `CAUSE_UNREPRESENTED`).
- **Refuted:** `retinoid_visual_cycle_disruption` as an anchor — its own scope is the RPE65/LRAT/ABCA4 chromophore relay; the protocols' retinoids are differentiation chemistry. No link.
- **Refuted:** a *new* organoid record — one already exists; augment instead.

8a. Schema-verification notes (iteration 4)

All identifiers and controlled values were checked against live schemas/ontologies:

- **Cell ontology IDs verified (OLS):** `CL:0000740` retinal ganglion cell ✓, `CL:0000604` retinal rod cell ✓, `CL:0000573` retinal cone cell ✓, `CL:0000210` photoreceptor cell ✓; `UBERON:0000966` retina (from the existing organoid record).
- **`ModelDivergenceTypeEnum` verified:** only four values exist — `BOUNDARY_OMISSION`, `PROXY_QUANTITY`, `CALIBRATION_PROVENANCE`, `CAUSE_UNREPRESENTED`. The report uses only valid ones.
- **`readouts` is structured, not free text:** range is `ExperimentalReadout` (slots: `name`, `target` [required, must repeat the link target], `phenotype_term`, `biomarker_term`, `biological_processes`, `assays`, `direction` [`ModelReadoutDirectionEnum`: INCREASED/DECREASED/UNCHANGED/RESTORED/ABOLISHED/ALTERED], `interpretation`, `evidence`). §4a was rewritten to this structure.
- **No duplicate records:** a full scan of `nams/` and `partial-nams/` finds only `retinal-organoid-pluripotent-stem-cell.yaml`; the RGC and photoreceptor records are genuinely new.
- **`resource: complement-arie-landscape-dashboard`** confirmed present at `resources/complement-arie-landscape-dashboard.yaml`.

9. Limitations & future directions

- No PRs were pushed from this run (no write access in-session); records and links above are complete drafts ready to open against nams-registry and dismech.
- NAMO `type`/architecture enum values for the two purified-neuron records are provisional and need confirmation against the imported NAMO schema.
- Primary-protocol DOIs (Sridhar, Luo = PMID ref 58, Rabesandratana, Ovando-Roche, Reichman) should be resolved and added as additional `references` when each protocol's specific data is curated onto a disease node.
